Euclidean Distance Geometry and the Augmented Lagrangian Method for Protein Structure Determination

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Abstract

Euclidean distance geometry (EDG) concerns the reconstruction of point coordinates from incomplete pairwise distances and has applications in protein structure determination. This paper examines the recovery of three-dimensional protein geometry from sparse inter-residue distances. We review the classical Lagrangian, penalty methods, and the augmented Lagrangian method, then formulate protein reconstruction using positive semidefinite Gram matrices. A factorized, gradient-based optimization approach is evaluated on a known protein structure under two levels of distance-constraint sparsity. The numerical results illustrate both successful reconstruction in the reported experiment and the sensitivity of recovery to the available geometric constraints.

1 Introduction: Euclidean Distance Geometry (EDG)

1.1 Motivation and implications

Protein folding is the process by which an amino-acid sequence gives rise to a protein's three-dimensional structure. Recovering this structure, even at the level of a residue-based point cloud, is important because geometry influences function. Binding interfaces, catalytic sites, stability, and conformational changes are all structural phenomena. Although modern deep-learning models can predict protein structures from sequences, a complementary mathematical question remains: how much distance information is required to recover a protein's three-dimensional geometry?

In experimental structural biology, distance information is often incomplete. For example, nuclear magnetic resonance (NMR) experiments can provide constraints on distances between selected atoms, rather than a complete matrix of all pairwise distances. This motivates the following reconstruction problem: given only a subset of inter-residue distances, can we recover the corresponding three-dimensional coordinates, up to geometric symmetries?

We approach this task by formulating coordinate recovery as an EDG problem. The protein is represented by n points in $\mathbb{R}^3$, such as the positions of the C_α atoms of its residues. Available distances become constraints on the unknown coordinates. Since these constraints are sparse, the problem may be underdetermined without additional geometric information. The central challenge is to reconstruct a globally consistent structure from a limited collection of pairwise distances.

More broadly, studying reconstruction from sparse measurements can help identify which geometric constraints are most informative and can guide the development of algorithms that remain stable when data are missing or noisy.

1.2 What is Euclidean distance geometry?

Euclidean distance geometry asks how much of a configuration of points is determined by some of its pairwise distances. Suppose $p_1, \ldots, p_n \in \mathbb{R}^d$ are unknown points. Rather than observing every distance $\|p_i - p_j\|_2$, we know the distances for only a subset of pairs. EDG seeks a configuration of points in $\mathbb{R}^d$ compatible with those observations.

1.3 EDG as a mathematical problem

Let

$$\Omega \subseteq \{(i,j) : 1 \le i < j \le n\}$$

be the set of observed pairs, with known squared distances $\{d_{ij}^2\}_{(i,j)\in\Omega}$. The EDG problem is to find $p_1, \ldots, p_n \in \mathbb{R}^d$ satisfying

$$\|p_i - p_j\|_2^2 = d_{ij}^2, \qquad (i,j) \in \Omega. \tag{1}$$

These constraints are quadratic in the unknown coordinates. Furthermore, distances are invariant under translations, rotations, and reflections, so a solution is not unique as a set of coordinates. Missing distances may introduce additional ambiguity.

A useful reformulation represents the points by a positive semidefinite Gram matrix. This makes the observed squared-distance constraints linear in the matrix entries, although enforcing a three-dimensional reconstruction still introduces a nonconvex rank restriction.

1.4 Challenges of EDG

Several features make EDG difficult in practice:

- **Nonconvexity:** In coordinate form, distance constraints are quadratic and the feasible set may be nonconvex.
- **Sparsity:** The observed pairs Ω may constitute only a small fraction of all possible pairs, leaving the problem underdetermined.
- **Invariances:** Multiple coordinate configurations related by rigid motions have identical pairwise distances.
- **Scalability:** Large structures require optimization over many coordinates or matrix entries.

These challenges motivate moving from point coordinates to Gram-matrix variables.

1.5 Paper overview

We first review equality-constrained optimization and the limitations of the classical Lagrangian and quadratic penalty methods. We then introduce the augmented Lagrangian method and apply it to a Gram-matrix formulation of protein reconstruction. Finally, we describe the gradient-based numerical method and examine reconstruction of a known protein using two sparse sets of distance constraints.

2 Lagrangians and Constrained Optimization

2.1 Constrained optimization background

Consider an equality-constrained problem

$$\min_{x \in \mathbb{R}^n} f(x) \qquad \text{subject to} \qquad c(x) = 0, \tag{2}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is the objective and $c : \mathbb{R}^n \to \mathbb{R}^m$ specifies the constraints. The feasible set is

$$\mathcal{F} = \{x \in \mathbb{R}^n : c(x) = 0\}.$$

Under suitable regularity conditions, this set may locally be a manifold. The optimization method must both reduce the objective and enforce feasibility.

2.2 The classical Lagrangian

The classical Lagrangian introduces multipliers $\lambda \in \mathbb{R}^m$:

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^\top c(x). \tag{3}$$

A constrained optimum can be studied through stationarity of the Lagrangian and satisfaction of the original constraints.

2.3 Optimality conditions

At a regular local solution (x^*, λ^*), the first-order necessary conditions are

$$\nabla_x \mathcal{L}(x^*, \lambda^*) = \nabla f(x^*) + J_c(x^*)^\top \lambda^* = 0, \qquad c(x^*) = 0, \tag{4}$$

where J_c is the Jacobian of c. Geometrically, the gradient of f at a constrained optimum is normal to the feasible manifold, rather than tangent to it. These conditions identify candidate stationary points but do not by themselves provide a numerical algorithm or guarantee a global minimum.

2.4 Limitations of the classical Lagrangian

The classical Lagrangian is useful for characterizing constrained optima, but directly solving its stationarity conditions can be numerically difficult. The associated system has a saddle-point structure; its Hessian may be indefinite, and standard unconstrained gradient descent does not directly enforce feasibility. This motivates methods with an explicit mechanism for penalizing constraint violations.

3 Penalty Methods and the Augmented Lagrangian Method

3.1 Penalty methods

A quadratic penalty method replaces the constrained problem by

$$\min_x \; f(x) + \frac{\rho}{2} \|c(x)\|_2^2, \qquad \rho > 0. \tag{5}$$

The second term penalizes constraint violations. As ρ increases, feasibility becomes more strongly favored. The objective can be treated using unconstrained optimization methods, but a very large penalty parameter may lead to poor numerical conditioning and slow convergence. A finite penalty does not generally enforce the constraints exactly.

3.2 Motivation for the augmented Lagrangian

Penalty methods trade stronger enforcement of constraints against numerical stability. The augmented Lagrangian method combines the quadratic penalty with Lagrange multipliers, using both constraint penalties and multiplier updates to guide the iterates toward feasibility.

3.3 Augmented Lagrangian method (ALM)

The augmented Lagrangian is

$$\mathcal{L}_\rho(x, \lambda) = f(x) + \lambda^\top c(x) + \frac{\rho}{2} \|c(x)\|_2^2 . \tag{6}$$

Its iterations approximately minimize $\mathcal{L}_\rho(x, \lambda^k)$ over x, then update the multipliers according to

$$\lambda^{k+1} = \lambda^k + \rho\, c(x^{k+1}). \tag{7}$$

This allows temporary constraint violations during the inner optimization while using the dual update to improve feasibility. Convergence depends on the properties of the problem and on how the iterations are implemented.

3.4 Comparison with previous approaches

Unlike classical stationarity equations, ALM provides a practical iterative scheme. Relative to a pure penalty method, multiplier updates can reduce the need for extremely large penalty parameters. This makes ALM attractive for constrained numerical problems, although individual inner subproblems may still be difficult and nonconvex.

4 A Gradient-Based ALM Algorithm for EDG

4.1 Why iterative methods are needed

EDG problems may involve high-dimensional variables, sparse distance measurements, and nonconvex constraints. Iterative first-order methods offer a practical compromise between computation time and reconstruction accuracy.

4.2 Gradient-based inner minimization

At iteration k, an approximate solution to

$$\min_x \mathcal{L}_\rho(x, \lambda^k)$$

is obtained using a gradient-based method. This avoids explicitly forming or inverting large Hessian matrices.

4.3 Overall algorithm structure

The method alternates between gradient-based minimization with respect to the primal variables and multiplier updates that penalize unsatisfied distance constraints. Stopping criteria can account for both progress in the objective and the norm of the constraint residual.

4.4 Practical considerations

Performance depends on the penalty parameter, the accuracy of inner minimization, the initialization, and the distribution of observed distances. Sparsity can reduce computational cost. Centering coordinates and accounting for rotational invariance also help remove redundant degrees of freedom when analyzing a reconstructed structure.

5 Protein Structure Recovery as a Euclidean Distance Geometry Problem

We now specialize EDG to residue-level protein reconstruction. Let the protein consist of n points $p_1, \dots, p_n \in \mathbb{R}^3$, representing, for example, the C_α atoms. For observed pairs $(i, j) \in \Omega$, the known squared distances are $d_{ij}^2 = \|p_i - p_j\|_2^2$. The objective is to recover coordinates consistent with these distances. As in (1), a direct coordinate formulation is nonlinear, motivating a Gram-matrix approach related to low-rank matrix completion [1].

5.1 From coordinates to a Gram matrix

Let $P \in \mathbb{R}^{n\times 3}$ contain the point coordinates as rows. Define the Gram matrix

$$X = PP^\top \in \mathbb{R}^{n\times n}. \tag{8}$$

Its entries are $X_{ij} = p_i^\top p_j$. By construction, X is symmetric, positive semidefinite ($X \succeq 0$), and has rank at most three.

5.2 Distances become linear constraints in the Gram matrix

The squared-distance identity is

$$\|p_i - p_j\|_2^2 = p_i^\top p_i + p_j^\top p_j - 2p_i^\top p_j = X_{ii} + X_{jj} - 2X_{ij}. \tag{9}$$

Consequently, each observed squared distance imposes a linear constraint on X:

$$d_{ij}^2 = X_{ii} + X_{jj} - 2X_{ij}, \qquad (i, j) \in \Omega.$$

5.3 Encoding the distance constraints

Write $\Omega = \{(i_1, j_1), \dots, (i_m, j_m)\}$, and define a linear measurement map $\mathcal{A} : \mathbb{R}^{n\times n} \to \mathbb{R}^m$ by

$$[\mathcal{A}(X)]_k = X_{i_k i_k} + X_{j_k j_k} - 2X_{i_k j_k}.$$

If $b = (d_{i_1 j_1}^2, \dots, d_{i_m j_m}^2)^\top$, the observations can be expressed compactly as

$$\mathcal{A}(X) = b. \tag{10}$$

5.4 Low-rank completion viewpoint

The EDG reconstruction problem becomes

$$\text{find } X \quad \text{such that} \quad \mathcal{A}(X) = b, \quad X \succeq 0, \quad \operatorname{rank}(X) \le 3. \tag{11}$$

The rank constraint encodes the requirement that the points lie in three-dimensional Euclidean space. Dropping it permits higher-dimensional embeddings: the observed distances may be satisfied in a larger space without producing an appropriate three-dimensional reconstruction. Promoting low rank is therefore central to the formulation.

5.5 Convex surrogate via nuclear-norm minimization

Direct rank constraints are nonconvex. A common surrogate minimizes the nuclear norm:

$$\min_X \|X\|_* \qquad \text{subject to} \qquad \mathcal{A}(X) = b, \quad X \succeq 0. \tag{12}$$

Since X is positive semidefinite, its nuclear norm equals the sum of its eigenvalues, or its trace:

$$\|X\|_* = \operatorname{tr}(X).$$

The resulting optimization problem is convex and favors low-nuclear-norm solutions. Nevertheless, this relaxation does not automatically ensure rank three or uniquely determine the physical structure.

5.6 Practical factorization used in the implementation

Rather than optimizing over the full matrix X, we use the factorization

$$X = PP^\top, \qquad P \in \mathbb{R}^{n\times r}.$$

This enforces $X \succeq 0$ and $\operatorname{rank}(X) \le r$ automatically. Moreover,

$$\operatorname{tr}(X) = \operatorname{tr}(PP^\top) = \|P\|_F^2 .$$

The factorized problem is therefore

$$\min_{P\in\mathbb{R}^{n\times r}} \|P\|_F^2 \qquad \text{subject to} \qquad \mathcal{A}(PP^\top) = b. \tag{13}$$

Although (12) is convex in X, the factorized formulation (13) is generally nonconvex in P.

5.7 Augmented Lagrangian formulation

Define the residual $c(P) = \mathcal{A}(PP^\top) - b$. The augmented Lagrangian is

$$\mathcal{L}_\rho(P,\lambda) = \|P\|_F^2 + \lambda^\top c(P) + \frac{\rho}{2}\|c(P)\|_2^2 . \tag{14}$$

With the scaled multiplier $u = \lambda/\rho$, completing the square gives, up to a term independent of P,

$$\|P\|_F^2 + \frac{\rho}{2}\|c(P) + u\|_2^2 .$$

This leads to the iterations

$$P^{k+1} \approx \arg\min_P \left\{ \|P\|_F^2 + \frac{\rho}{2}\left\|\mathcal{A}(PP^\top) - b + u^k\right\|_2^2 \right\}, \tag{15}$$

$$u^{k+1} = u^k + \mathcal{A}(P^{k+1}(P^{k+1})^\top) - b. \tag{16}$$

Both the penalty parameter ρ and the working rank r affect optimization. A choice $r = 3$ restricts the solution to three dimensions but may make the search more difficult. Allowing $r > 3$ provides additional freedom during optimization, after which a three-dimensional representation can be extracted. A larger working rank does not guarantee that the leading three-dimensional reconstruction preserves all the fitted distances; its accuracy must be assessed after extraction.

5.8 Extraction of three-dimensional coordinates

After optimization, form $X = PP^\top$. To remove translation ambiguity, apply the centering operator

$$J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top, \qquad X_c = JXJ.$$

Compute an eigendecomposition

$$X_c = V\Lambda V^\top,$$

where eigenvalues are ordered from largest to smallest. Retain the three leading nonnegative eigenvalues and eigenvectors to obtain

$$Y = V_{[:,1:3]}\Lambda_{1:3}^{1/2} \in \mathbb{R}^{n\times 3}. \tag{17}$$

The Gram matrix $YY^\top$ is the best rank-three approximation to X_c in Frobenius norm. These coordinates can then be compared with the known reference geometry after alignment.

6 Applying the Method in Code and Results

We tested the method using a previously determined protein structure, PDB 1IXC, and asked whether optimization from a random initial point cloud could reconstruct the reference coordinates from a sparse subset of pairwise distances.

For the first experiment, distances exceeding 9 Å were discarded, retaining approximately 4.3% of the possible pairwise constraints. These distances were calculated from the known structure and selectively retained to simulate incomplete information; they were not separately measured in this experiment. The sparsity pattern is shown in Figure 1.

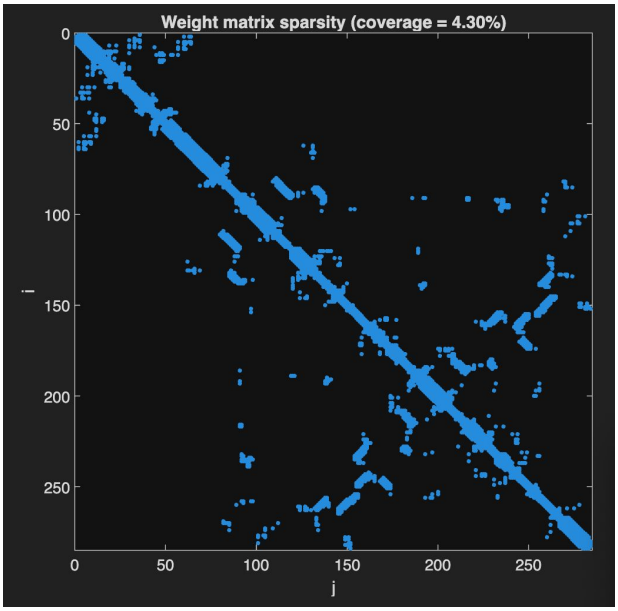
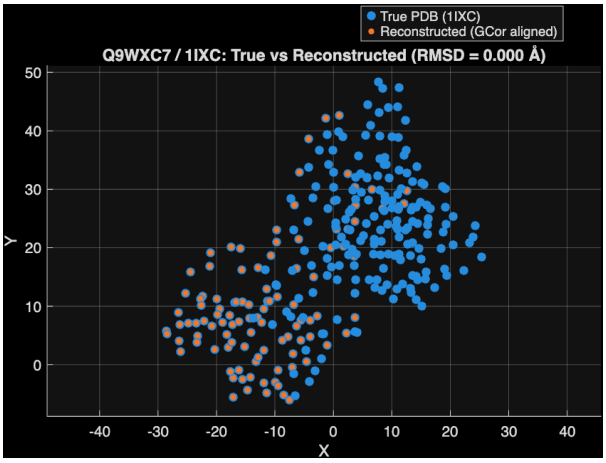


Figure 1: Sparsity pattern of the distance-constraint matrix for the 9 Å cutoff.

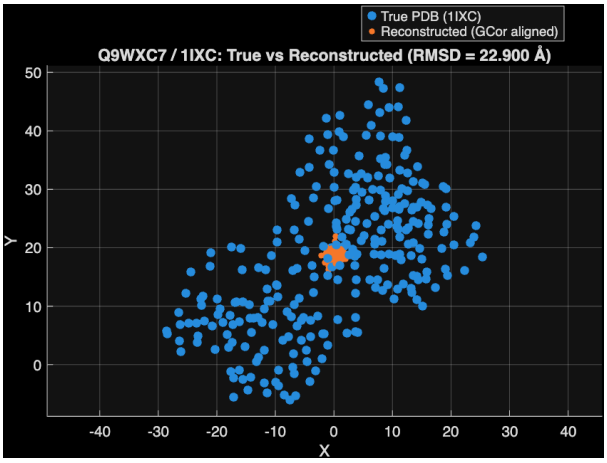
The resulting constrained optimization problem was solved using the augmented Lagrangian formulation, with Barzilai–Borwein gradient-descent steps for the inner minimization. After optimization, we extracted three-dimensional coordinates as described above. The reconstructed

coordinates were aligned with the true structure using Procrustes analysis, which finds a rigid alignment that minimizes the sum of squared deviations between corresponding points.

Figure 2 shows the original visual comparisons. At 4.3% coverage the experiment reported an RMSD of 0.000 Å after alignment. When the cutoff was lowered to 6 Å, the retained constraints decreased to approximately 1.9% of all pairs and the reported RMSD increased to 22.900 Å.



(a) 9 Å cutoff: 4.3% constraint coverage.



(b) 6 Å cutoff: 1.9% constraint coverage.

Figure 2: Reference and reconstructed protein structures after Procrustes alignment, as shown in the original computational results. The plots display the RMSD values reported for each experiment.

The difference between these experiments suggests that reconstruction can be sensitive to the available distances. In particular, reducing the distance cutoff changes not only the number of observed pairs but also which parts of the structure are linked by constraints. The experiments therefore motivate several questions. Is failure at lower coverage caused mainly by the number of constraints, or by the loss of longer-range connections between residues? Does the required amount of information depend on the protein's geometry? How does initialization affect convergence? Answering these questions would require systematic experiments across multiple proteins and constraint-selection strategies.

Works Cited

[1] Abiy Tasissa and Rongjie Lai. "Exact Reconstruction of Euclidean Distance Geometry Problem Using Low-rank Matrix Completion." *arXiv:1804.04310*, 2018.